Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 3
VRAESTEL

Semester Test 3
QUESTION PAPER

Module EBN111
Elektrisiteit en Elektronika
Mei 2018

Module EBN111
Electricity and Electronics
May 2018

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

MEMO.

Eksamensinligting: *Exam information:*			
Maksimum punte: *Maximum marks:*	46	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 min + 10 min to complete OCR form *90 min + 10 min om OKH vorm te voltooi*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		17 + 1 Answer sheets	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die ANTWOORDBLAD en op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
   All questions must be answered on the ANSWER SHEET and on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. D Ramotsoela
	3. X Ye

STEPS TO FOLLOW

Do step 0 before the start of the session.
Do step 1 in the first 90 minutes of the test.
Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min):

- Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test.
- Write down the answers for each question on the PRACTICE ANSWER SHEET at the end of the QUESTION PAPER. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min):

Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2018EBN111S3**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- Do not use commas! Please make use of full stops.

- Complex numbers:

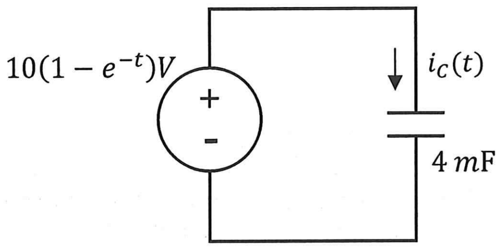
Rectangular form: always place the " j " **at the start** of the imaginary part of the complex number.
Polar form: always add the " ° " (degree sign) after the angle, and make a clear " ∠ " (angle sign).

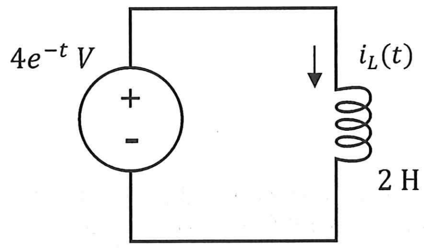
Examples

01	–	1	5	4	.	9				k	Ω							
02	1	.	2							r	a	d	÷	s				
03	–	1	.	4	+	j	8			µ	A							
04	1	4	0	°														
05	3	4	0	∠	4	5	°			V								
06	–	j	5	4						k	Ω							
07	1	∠	2	0	°													
08	5	0	∠	–	1	3	0	°		m	A							

SECTION / AFDELING 1 [10]

Questions 1-2 [4]		Vraag 1-2 [4]	
It is known that $i_C(0) = 40\ mA$ for the circuit with the capacitor, and $i_L(0) = 0\ A$ for the circuit with the inductor. Calculate $i_C(t)$ and $i_L(t)$ at $t = 1\ s$.		Dit is bekend dat $i_C(0) = 40\ mA$ vir die kringbaan met die kapasitor en $i_L(0) = 0\ A$ vir die kringbaan met die induktor. Bereken $i_C(t)$ en $i_L(t)$ vir $t = 1\ s$.	
01	$i_C(1\ s)$	**01**	$i_C(1\ s)$
02	$i_L(1\ s)$	**02**	$i_L(1\ s)$





$*\ i_c(0) = 40mA$

$i_c(t) = C\frac{dv_c}{dt}$

$= 4m\frac{d}{dt}(10 - 10e^{-t})$

$= +40me^{-t}$

$i_c(1) = 0.04e^{-1}$

$= 14.715\ mA.$

$*\ i_L(0) = 0A.$

$v_L(t) = L\frac{di_L}{dt}.$

$\therefore i_L(t) = \frac{1}{L}\int_0^t v_L\,dt + i_L(0)$

$= \frac{1}{2}\int_0^t 4e^{-t}\,dt$

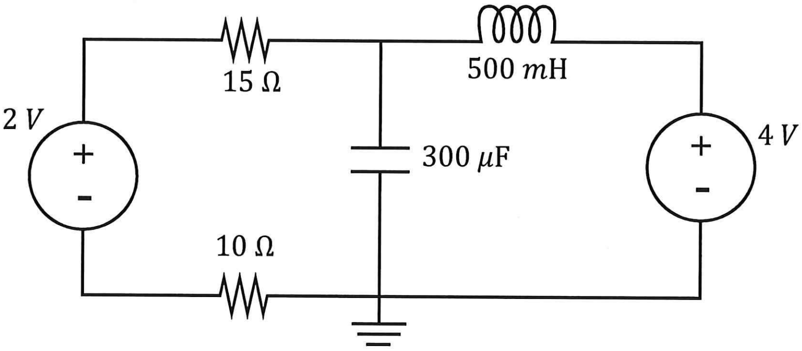
$= -2e^{-t}\Big|_0^t$

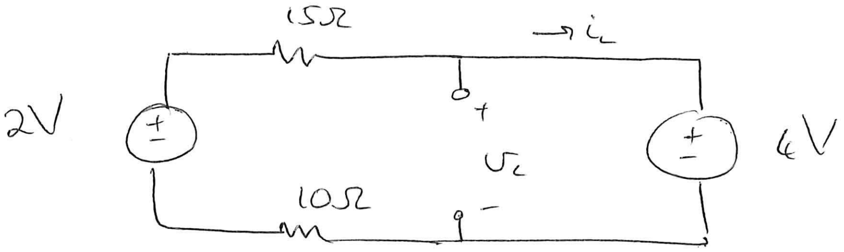
$= -2e^{-t} + 2.$

$i_L(1) = 2 - 2e^{-1}$

$= 1.264\ A$

Questions 3-4 [4]		Vraag 3-4 [4]	
Assume steady-state conditions (Direct Current) for the circuit below. What is the energy stored by the capacitor w_C and inductor w_L?		Aanvaar gestadigde toestand (Gelykstroom) vir onderstaande kringbaan. Wat is die energie gestoor deur die kapasitor w_C en induktor w_L?	
03	w_C	**03**	w_C
04	w_L	**04**	w_L





$$-2 + 15 i_L + 4 + 10 i_L = 0$$

$$i_L = \frac{-2}{25} = -80\,mA.$$

$$v_c = 4V$$

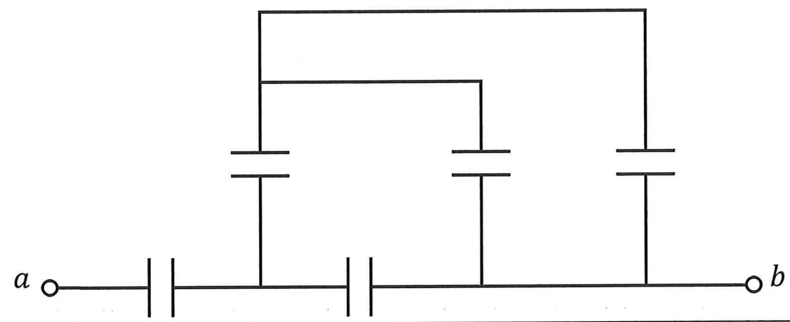
$$\therefore w_L = \frac{1}{2}(0.5)(-80m)^2$$

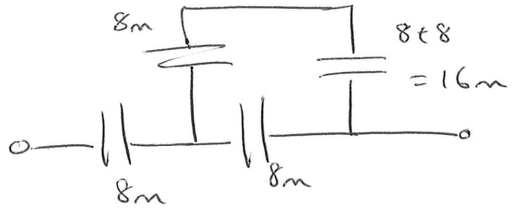
$$= \underline{1.6\,mJ}$$

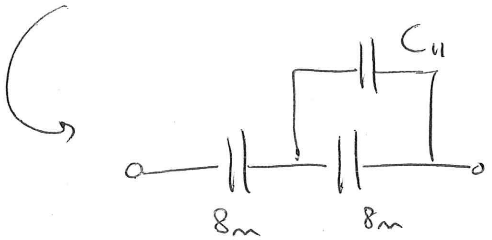
$$w_c = \frac{1}{2}(300\mu)(4)^2$$

$$= \underline{2.4\,mJ}$$

Question 5 [2]		Vraag 5 [2]	
05	Determine C_{ab} if all capacitors are 8 mF.	**05**	Bepaal C_{ab} indien alle kapasitors 8 mF.





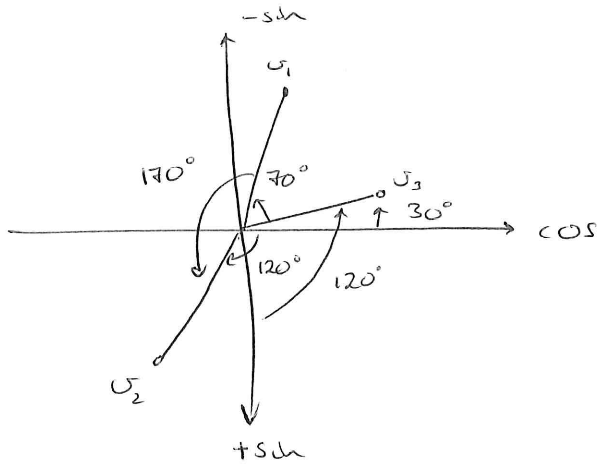


$C_{11} = 8m \,||\, 16m$
$= 5.333\,mF$

$C_{ab} = 8m \,||\, (5.333m + 8m)$
$= 8m \,||\, (13.333m)$
$= \underline{5\,mF}$ ✓✓

SECTION/AFDELING 2 [18]

Questions 6-7 [4]		Vraag 6-7 [4]	
Three sinusoid signals are given. $v_1 = 10\cos(\omega t + 70°)$ $v_2 = 6\cos(\omega t - 120°)$ $v_3 = 12\sin(\omega t + 120°)$		Drie sinusgolwe word gegee: $v_1 = 10\cos(\omega t + 70°)$ $v_2 = 6\cos(\omega t - 120°)$ $v_3 = 12\sin(\omega t + 120°)$	
06	Which statement is correct (write 1, 2, 3 or 4)? 1. v_1 leads v_2 by 170 ° and leads v_3 by 40 °. 2. v_1 lags v_2 by 170 ° and leads v_3 by 110 °. 3. v_1 lags v_2 by 170 ° and leads v_3 by 40 °. 4. v_1 leads v_2 by 170 ° and lags v_3 by 110 °.	**06**	Watter stelling is korrek (skryf 1, 2, 3, of 4)? 1. v_1 loop voor v_2 met 170 ° en na v_3 met 40 °. 2. v_1 loop na v_2 met 170 ° en voor v_3 met 110 °. 3. v_1 loop na v_2 met 170 ° en voor v_3 met 40 °. 4. v_1 loop voor v_2 met 170 ° en na v_3 met 110 °.
07	Determine: $\bar{V}_T = \bar{V}_1 + \bar{V}_2 - \bar{V}_3$ and write the final answer in ***polar form***.	**07**	Bepaal: $\bar{V}_T = \bar{V}_1 + \bar{V}_2 - \bar{V}_3$ en skryf die finale antwoord in ***polêre vorm***.

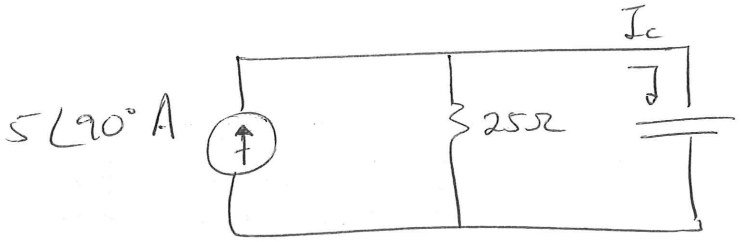


* Statement # 3 ✓✓

$$\bar{V}_T = 10\angle 70° + 6\angle -120° - 12\angle 30°$$

$$= \underline{10.133 \angle -169.8°}$$

Question 8 [2]	Vraag 8 [2]
The following three elements are all connected in parallel: a $5\cos(80t + 90°)\ A$ current source, a 25 Ω resistor, and a 250 μF capacitor. What is the current through the capacitor? (Assume the positive form of Ohm's Law.)	Die volgende drie elemente word in parallel gekoppel: 'n $5\cos(80t + 90°)\ A$ stroombron, 'n 25 Ω resistor, en 'n 250 μF kapasitor. Wat is stroom deur die kapasitor? (Aanvaar die positiewe vorm van Ohm se Wet.)
08 $\bar{I}_C$ **(Rectangular format)**	**08** $\bar{I}_C$ **(Reghoekige formaat)**



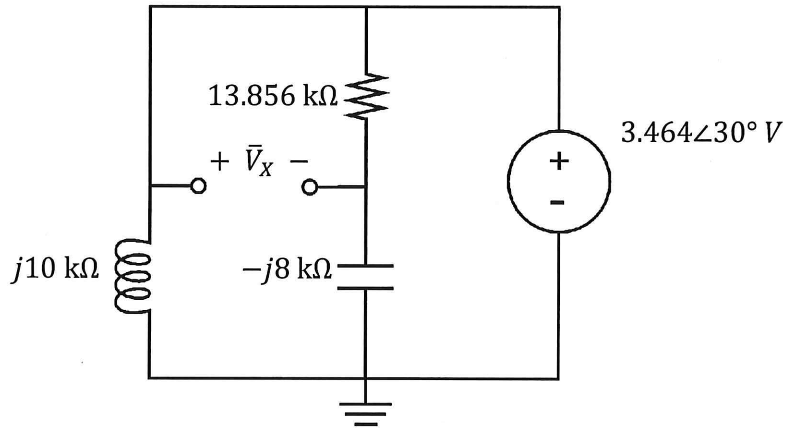
$$Z_C = -j/_{80\times 250\mu}$$

$$= -j50\ \Omega$$

$$\bar{I}_C = \frac{j5\times 25}{25-j50} \quad \text{(I-division)}$$

$$= -2+j\ \text{A}$$ ✓✓

Question 9 [2]		Vraag 9 [2]	
09	Calculate $\bar{V}_X$. **(Polar format)**	09	Bereken $\bar{V}_X$. **(Polêre formaat)**

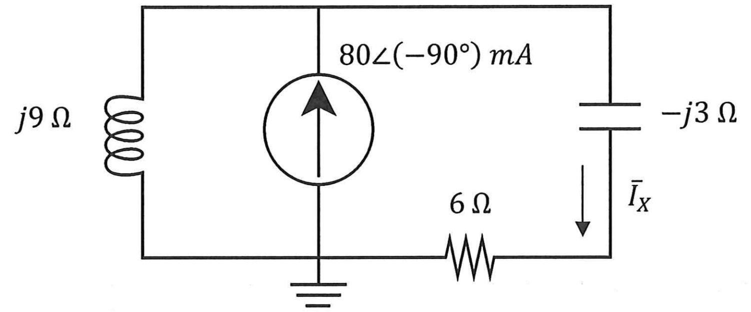


$$\bar{V}_x = 3.464\angle 30^\circ - \left(\frac{3.464\angle 30^\circ \times (-j8k)}{13.856k - j8k}\right)$$

$$= 3.464\angle 30^\circ - 1.732\angle -30^\circ$$

$$= \underline{3\angle 60^\circ\ V}$$

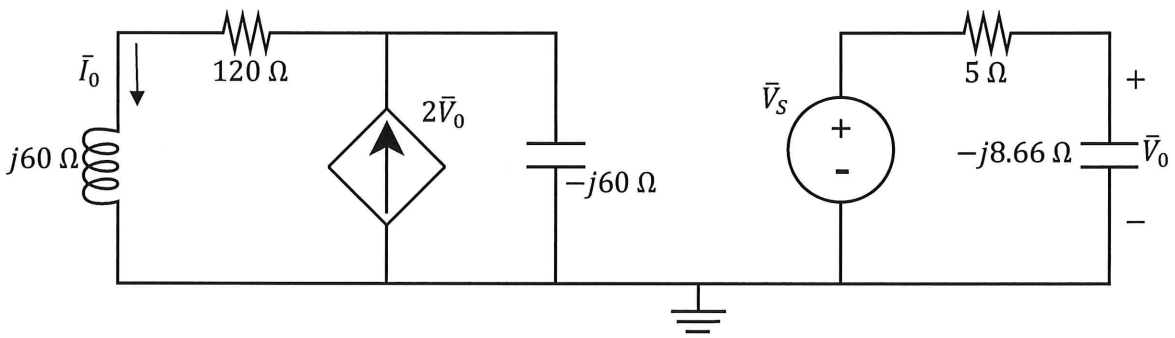
Question 10 [2]		Vraag 10 [2]	
10	Calculate $\bar{I}_X$. **(Rectangular format)**	10	Bereken $\bar{I}_X$. **(Reghoekige formaat)**



$$I_x = \frac{-j80\,mA \times j9}{j9 - j3 + 6} \quad (I\text{-division})$$

$$= \underline{60 - j60\ mA}$$ ✓✓

Question 11-12 [4]		Vraag 11-12 [4]	
If $\bar{V}_0 = 1.732\angle(60°)\ V$, calculate $\bar{V}_S$ and $\bar{I}_0$.		Indien $\bar{V}_0 = 1.732\angle(60°)\ V$, bereken $\bar{V}_S$ en $\bar{I}_0$.	
11	$\bar{V}_S$ **(Polar format)**	**11**	$\bar{V}_S$ **(Polar format)**
12	$\bar{I}_0$ **(Polêre formaat)**	**12**	$\bar{I}_0$ **(Polêre formaat)**



$$\bar{V}_o = \frac{-j8.66 \times \bar{V}_s}{5 - j8.66}$$ (V-division)

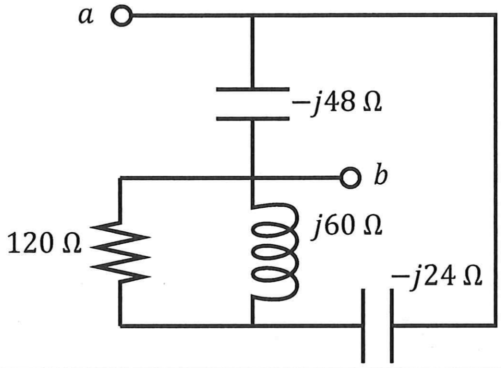
$$\therefore \bar{V}_s = \frac{1.732\angle 60° \times (5 - j8.66)}{-j8.66}$$

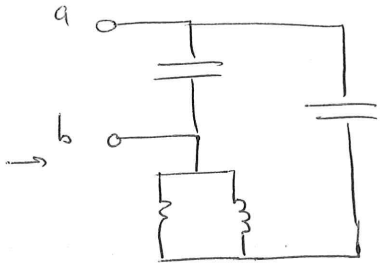
$$= \underline{2\angle 90°\ V}$$

$$\bar{I}_o = \frac{(2\bar{V}_o)(-j60)}{-j60 + j60 + 120}$$

$$= \underline{1.732\angle -30°\ A}$$

Question 13 [2]		Vraag 13 [2]	
13	Calculate Z_{ab}. **(Polar format)**	13	Bereken Z_{ab}. **(Polêre formaat)**





$$Z_{ab} = (120 \| j60 - j24) \| (-j48)$$

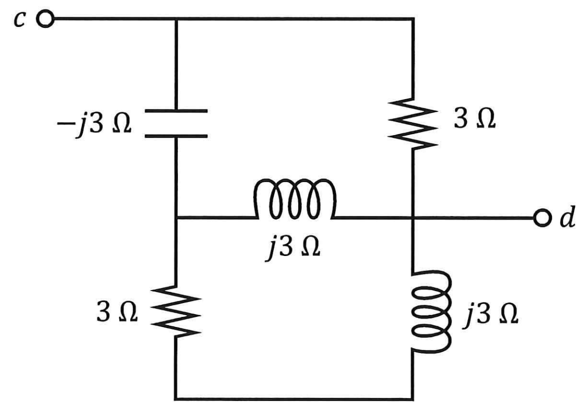
$$= \left(\frac{120 \times j60}{120 + j60} - j24\right) \| (-j48)$$

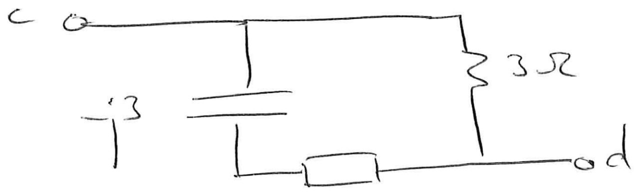
$$= (24 + j48 - j24) \| (-j48)$$

$$= \frac{(24 + j24)(-j48)}{24 + j24 - j48}$$

$$= \underline{48\ \Omega}$$

Question 14 [2]		Vraag 14 [2]	
14	Calculate Z_{cd}. **(Rectangular format)**	14	Bereken Z_{cd}. **(Reghoekige formaat)**





$$Z_1 = (3+j3) \| j3$$

$$= \frac{(3+j3)(j3)}{3+j6} = 0.6 + j1.8\ \Omega$$

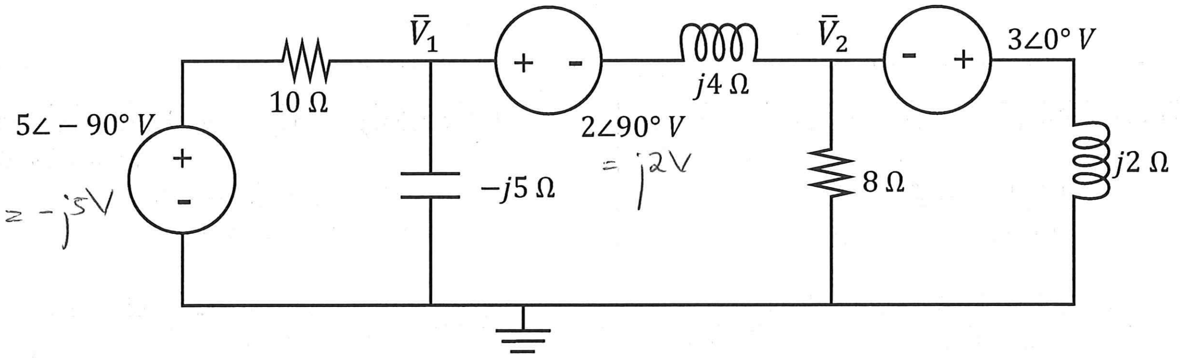
$$Z_{cd} = (0.6 + j1.8 - j3) \| 3$$

$$= \frac{(0.6 - j1.2)(3)}{3.6 - j1.2}$$

$$= \underline{0.75 - j\,0.75\ \Omega}$$

SECTION / AFDELING 3 [18]

Question 15-18 [4]							
Use **nodal** analysis to set up two equations of the following form: $$a\bar{V}_1 + b\bar{V}_2 = 10 + j10$$ $$c\bar{V}_1 + d\bar{V}_2 = -12 - j4$$ (Give the coefficients in **rectangular format.**)				Gebruik **node** analise om twee vergelykings van die volgende vorm op te stel: $$a\bar{V}_1 + b\bar{V}_2 = 10 + j10$$ $$c\bar{V}_1 + d\bar{V}_2 = -12 - j4$$ (Gee die koeffisiente in **reghoekige formaat.**)			
15	a	**16**	b	**15**	a	**16**	b
17	c	**18**	d	**17**	c	**18**	d



① $$\frac{V_1 + j5}{10} + \frac{V_1}{-j5} + \frac{V_1 - j2 - V_2}{j4} = 0$$

$(\times j20)$ $$j2V_1 + j^2 10 - 4V_1 + 5V_1 - j10 - 5V_2 = 0$$

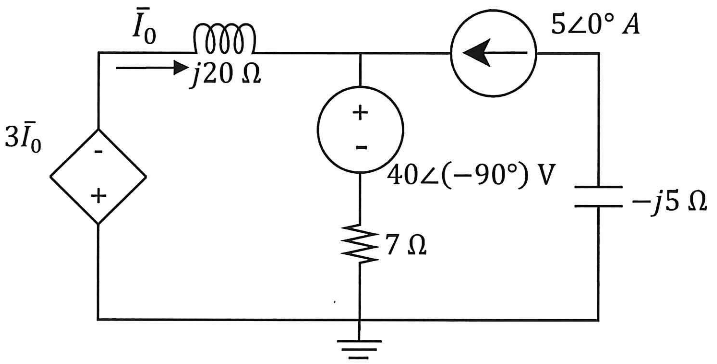
$$\underline{V_1(1 + j2) - 5V_2 = 10 + j10}$$

② $$\frac{V_2 + j2 - V_1}{j4} + \frac{V_2}{8} + \frac{V_2 + 3}{j2} = 0$$

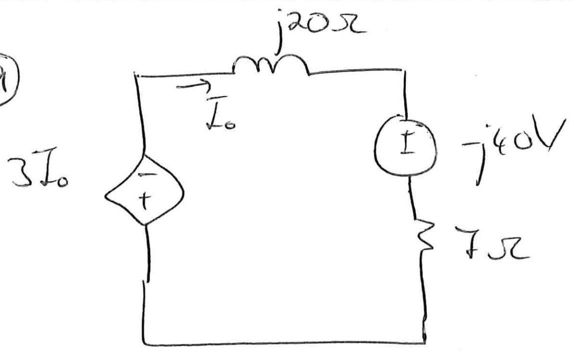
$(\times j8)$ $$2V_2 + j4 - 2V_1 + jV_2 + 4V_2 + 12 = 0$$

$$\underline{-2V_1 + V_2(6 + j) = -12 - j4}$$

Question 19-20 [4]			
a) What is the unique contribution of the independent **voltage source** to $\bar{I}_0$? b) What is the unique contribution of the independent **current source** to $\bar{I}_0$?		a) Wat is die unieke bydrae van die onafhanklike **spanningsbron** tot $\bar{I}_0$? b) Wat is die unieke bydra van die onafhanklike **stroombron** tot $\bar{I}_0$?	
19	a) $\bar{I}_{0(V)}$ **(Rectangular format)**	**19**	a) $\bar{I}_{0(V)}$ **(Reghoekige formaat)**
20	b) $\bar{I}_{0(A)}$ **(Rectangular format)**	**20**	b) $\bar{I}_{0(A)}$ **(Reghoekige formaat)**



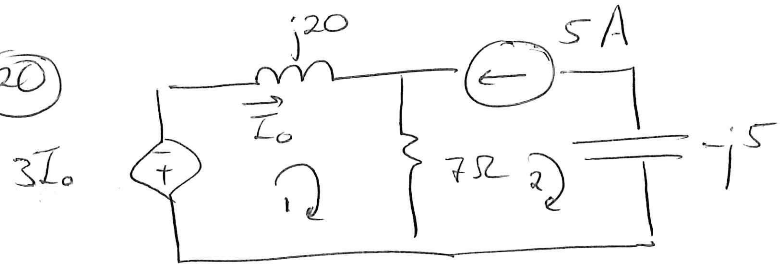
19



$$+3I_o + j20I_o - j40 + 7I_o = 0.$$

$$I_o = \frac{j40}{10+j20} = \underline{1.6 + j0.8\ A}$$

20



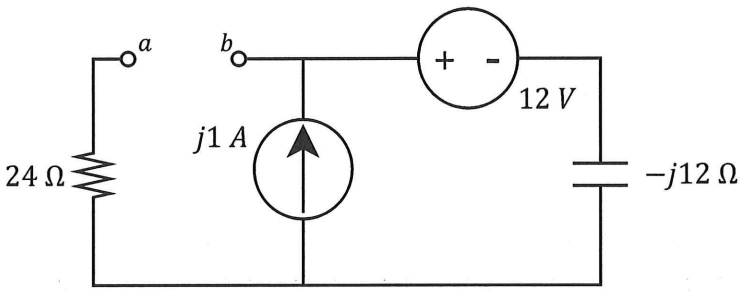
* $I_2 = -5A.$ * $I_o = I_1$

* $+3I_o + j20I_o + 7I_o - 7I_2 = 0.$

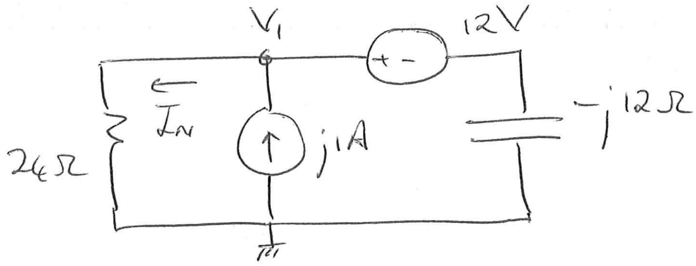
$$I_o(10 + j20) = 7(-5)$$

$$I_o = \underline{-0.7 + j1.4\ A}$$

Question 21-22 [4]					
Determine the Norton equivalent w.r.t. terminals *a-b*.			Bepaal die Norton ekwivalent m.v.t. terminale *a-b*.		
21	$\bar{I}_N$	**(Rectangular format)**	**21**	$\bar{I}_N$	**(Reghoekige formaat)**
22	Z_N	**(Rectangular format)**	**22**	Z_N	**(Reghoekige formaat)**



$\underline{\underline{I_N}}$



$$\frac{V_1}{24} + \frac{V_1 - 12}{-j12} = j1$$

$$(\times j24) \quad jV_1 - 2V_1 + 24 = -24.$$

$$V_1 = \frac{-48}{-2+j} = 19.2 + j9.6\,V$$

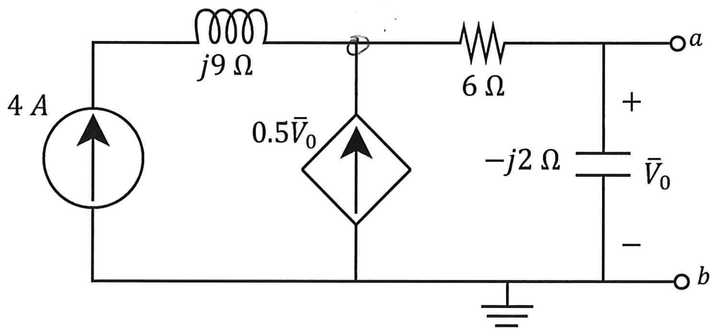
$$\therefore I_N = \frac{V_1}{24} = \underline{800 + j400\ mA}$$

Current flows from a to b. Thus I_N = -800 - j400 mA.

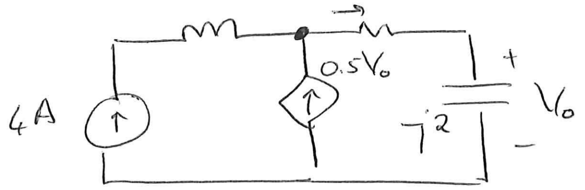
$\underline{\underline{Z_N}}$

$$Z_N = \underline{24 - j12\ \Omega}$$

Question 23-24 [4]					
Determine the Thevenin equivalent w.r.t. terminals *a-b*.			Bepaal die Thevenin ekwivalent m.v.t. terminale *a-b*.		
23	$\bar{V}_{TH}$	**(Polar format)**	**23**	$\bar{V}_{TH}$	**(Polêre formaat)**
24	Z_{TH}	**(Rectangular format)**	**24**	Z_{TH}	**(Reghoekige formaat)**



V_{TH}

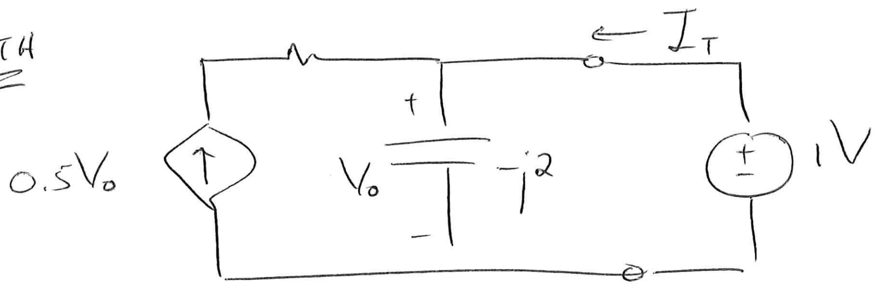


$$4 + 0.5V_o = \frac{V_o}{-j2}$$

$$\therefore -j8 - jV_o = V_o \quad \checkmark\checkmark$$

$$V_o = \frac{-j8}{1+j} = \underline{5.657\angle -135^\circ \text{ V}}$$

Z_{TH}



$$V_o = 1V$$

$$\therefore I_T = -0.5V_o + \frac{V_o}{-j2} = -0.5 + j0.5$$

$$\therefore Z_{TH} = \frac{1}{-0.5 + j0.5} = \underline{-1 - j1\ \Omega} \quad \checkmark\checkmark$$

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	8	E	B	N	1	1	1	S	3		

Answer — Unit

01	$i_C(1s)$ = 14.715	mA
02	$i_L(1s)$ = 1.264	A
03	w_C = 2.4	mJ
04	w_L = 1.6	mJ
05	C_{ab} = 5	mF
06	*Statement #:* 3	*No unit required.*
07	$\bar{V}_T$ = 10.133 ∠−169.8° ≃ 10∠−170° (Polar)	
08	$\bar{I}_C$ = −2 + j (Rectangular)	A
09	$\bar{V}_X$ = 3∠60° (Polar)	V
10	$\bar{I}_X$ = 60 − j60 (Rectangular)	mA
11	$\bar{V}_S$ = 2∠90° (Polar)	V
12	$\bar{I}_0$ = 1.732 ∠−30° (Polar)	A
13	Z_{ab} = 48 (Polar)	Ω
14	Z_{cd} = 0.75 − j0.75 (Rectangular)	Ω
15	a = 1 + j2 (Rectangular)	*No unit required.*
16	b = −5 (Rectangular)	
17	c = −2 (Rectangular)	
18	d = 6 + j (Rectangular)	
19	$\bar{I}_{0(V)}$ = 1.6 + j0.8 (Rectangular)	A.
20	$\bar{I}_{0(A)}$ = −0.7 + j1.4 (Rectangular)	A
21	$\bar{I}_N$ = ~~800 + j400~~ -0.8 - j0.4 A (Rectangular)	mA
22	Z_N = 26 − j12 (Rectangular)	Ω
23	$\bar{V}_{TH}$ = 5.657 ∠−135° (Polar)	V
24	Z_{TH} = −1 − j (Polar)	Ω